

Psychiatry

World of Revision

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Contents

Psychiatry

Psychosis Related Disorders	1
Psychotic disorders • Schizophrenia : Introduction • Schizophrenia : C/F • Schizophrenia : Prognostic factors • Antipsychotics • Extrapramidal symptoms (EPS)	
Mood Related Disorders	9
Depression • Bipolar disorder • Persistent mood disorders • Postpartum mood disorders • Suicidal behaviour • Treatment resistant depression • Pharmacological Management of bipolar disorder • Lithium therapy	
Anxiety, Stress and Other Disorders	17
Anxiety disorders • Panic attack vs. panic disorders • Phobia • Grief • Traumatic disorders • Obsessive, compulsive & related disorders • Somatization, dissociation and conversion disorders • Factitious disorders	
Eating, Sleep and Other Disorders	24
Eating disorders • EEG waves • Sleep disorders • Personality disorders • Sexual disorders • Substance use disorders	
Neurocognitive, Neurodevelopmental and Other Disorders	33
Delirium • Dementia • Neurodevelopmental disorders • Concepts related to mind • Mental health care act (MHCA) 2017	

PSYCHOSIS RELATED DISORDERS

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Psychotic Disorders

00:00:36

- Loss of touch with reality.
- marked disturbance in :
 - Thoughts (**Disorganized**, illogical or fragmented).
 - Perception (Hallucinations).
 - Emotions (Blunted, inappropriate or labile affect).
 - Behaviour (Disorganized or **catatonic** actions).
- E.g. : Schizophrenia, delusional disorder.

Psychotic vs. Neurotic Disorders :

Neurotic disorders :

- Generalized anxiety disorder
- Obsessive compulsive disorder

c/f : **JIPR** (mnemonic)

- **J**udgement
- **I**nsight
- **P**ersonality
- **R**eality

← ⊕

→ ⊖/↓

Psychotic disorders :

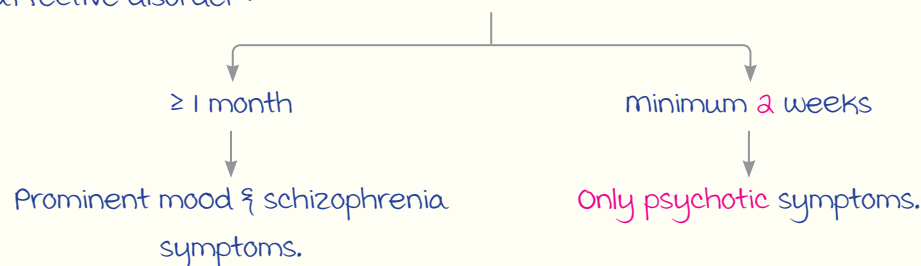
- Schizophrenia.
- Delusional disorders.

Timelines for Diagnosis :

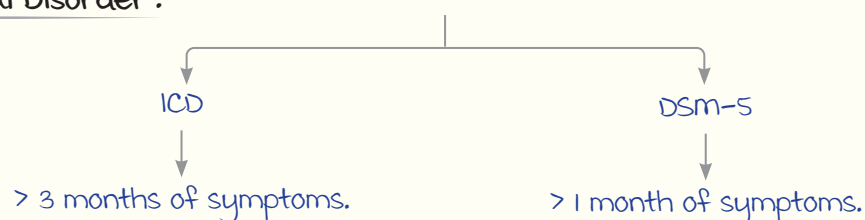
Psychotic disorders :

Duration	ICD	DSM-5
< 1 month	Acute & transient psychosis	Brief psychotic disorder
1 - 6 months	Schizophrenia	Schizophreniform illness
> 6 months		Schizophrenia

Schizoaffective disorder :



Delusional Disorder :



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Schizophrenia : Introduction

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Bleuler :

- Coined the term 'schizophrenia'.
- 4 A's :

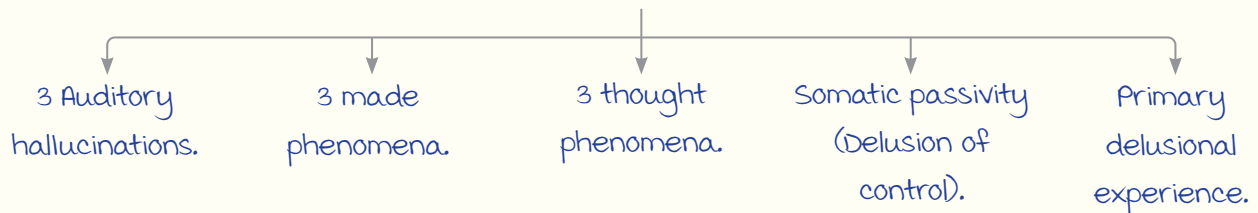


Note :

Auditory hallucination is not a part of 4 A's.

Positive symptoms :

Kurt Schneider's II first rank Symptoms :

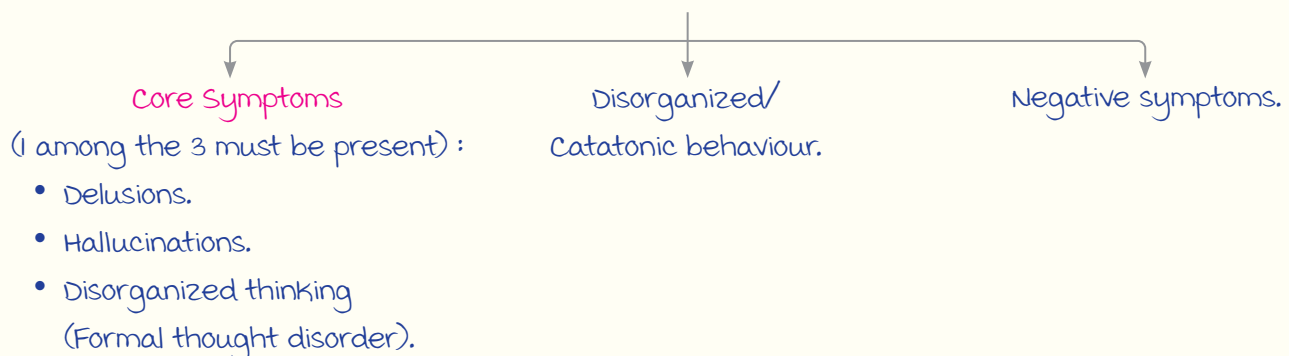


Negative symptoms :

- Apathy.
- Avolition.
- Affective flattening.
- Attention deficit.
- Anhedonia :
 - Loss of pleasure in previously pleasurable activities.
 - Also seen in depression.
- Alogia.

Diagnosis of Schizophrenia :

At least 2 of the following symptoms > 1 month (ICD) / > 6 months (DSM)



Schizophrenia : C/F

00:11:46

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Delusions :

- False & fixed belief.
- m/c delusion : Paranoid/Persecutory delusion.

Note :

Idea : False & fluctuating belief.

Named delusions :

		Features
Othello syndrome		• Delusional jealousy/delusion of infidelity • Common in chronic alcoholics
De Clerambault syndrome		erotomania/delusion of love
Cotard syndrome/ Walking corpse syndrome		Severe depression + nihilistic delusion (Delusion of negation)
magnan syndrome		• AKA cocaine psychosis/cocaine bugs/formication • Delusions of persecution + tactile hallucination (Insects crawling under skin)
Ekbom syndrome 🧠		• Delusional parasitosis • match box sign • Type of restless leg syndrome
Delusions of misidentification	Capgras syndrome	• Known person → Stranger • Negative misidentification/delusion of doubles
	Fregoli syndrome	• Stranger → Persecutor • Positive misidentification

Delusional disorder vs. Schizophrenia :

	Delusional disorder	Schizophrenia
Core symptom	Delusions	Delusions + Other psychotic symptoms
Delusion type	Simple	Complex & bizarre
Vegetative symptoms (Sleep & appetite)	Normal	Affected
Daily functioning	Normal	Affected

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Hallucinations :

Types :

	m/c associated disorder 
Visual	Organic mental illness (Neurocognitive disorders)
Auditory	Schizophrenia
Tactile	magnan syndrome

- True hallucination : Originates from **outer objective** space.
- Pseudo hallucination : Originates from **inner subjective** space.

Formal Thought Disorder :

- Disturbance in healthy thinking (Constancy + continuity + organization).
- AKA disorder of thought form/disorganized thinking/**word salad** (Verbigeration).

Features  :

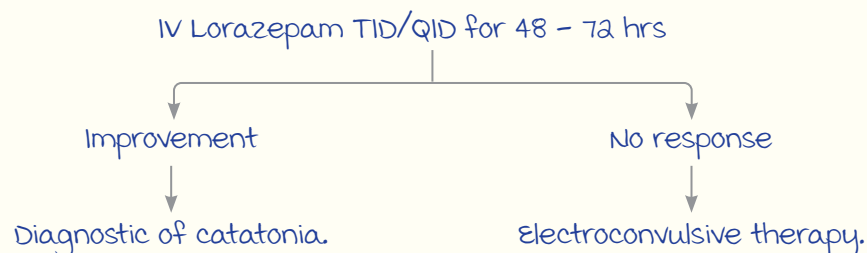
- Derailment.
- Tangentiality.
- **Neologism**.
- **Syllogistic thinking**.
- Autistic thinking.

Catatonia :Etiology : **Frontal lobe** dysfunction.

- mood disorders (mania > Depression).
- Schizophrenia.
- Organic/Neurocognitive conditions.

Clinical features  :

- **Psychological pillow** : Neck remains elevated even upon removal of pillow.
- Automatic obedience.
- **Catalepsy** : maintain same posture for a long time.
- **Waxy flexibility**.

Treatment  :**Lorazepam trial test :**

Schizophrenia : Prognostic Factors

00:31:20

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		Good factors	Bad factors
Gender 		Female	male
Onset		Acute	Young & insidious
Precipitating factors		⊕	⊖
Premorbid functioning		Well adjusted	Poor
mood symptoms		⊕	⊖
Family h/o	Schizophrenia	-	⊕
	mood disorder	⊕	-
Compliance to medication		⊕	⊖
Family & emotional support		⊕	⊖
Symptoms		more positive	more negative
Neurological signs/symptoms		⊖	⊕
H/o perinatal trauma		⊖	⊕
H/o assaultive behaviour		⊖	⊕
Relapses		-	multiple

Antipsychotics

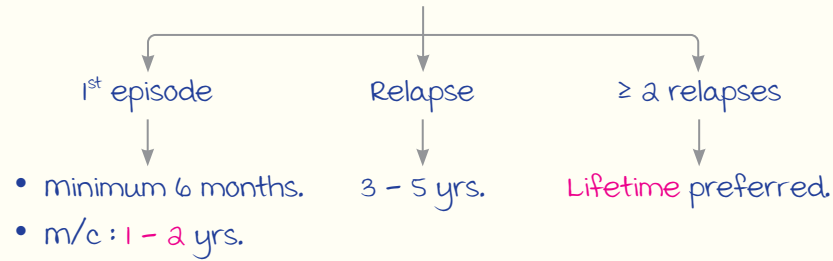
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Types :

Generation	Key points
First	Neurotoxicity
Second	metabolic toxicity
Third	• Partial agonist at D_2 & $5-HT_{1A}$ receptor • Examples : ABC. - Aripiprazole - Brexpiprazole - Cariprazine
Newer antipsychotics	• Pimavanserin : - Inverse agonist (At $5HT_{2A}$) - Rx of Parkinson's disease • Xanomeline + Trospium : - Act on muscarinic system

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Duration of treatment :

Clozapine :

Advantages	Disadvantages
• most effective • minimal EPS • minimal hyperprolactinemia	Severe s/e : • myocarditis • Agranulocytosis (~ 1% cases)
	Other s/e : • Weight gain • Sedation • Hyperglycemia • Dyslipidemia • Sialorrhea • ↓ Seizure threshold (Dose dependent risk)

monitoring for agranulocytosis :

- Agranulocytosis : Absolute neutrophil count (ANC) < 500.
- Regular blood tests (Total & differential WBC count) :



Indication : DOC for treatment resistant schizophrenia 
 (Refractory to two different antipsychotics).

Effects of Antipsychotics  :

Drug	Effect
Quetiapine	Cataract
Risperidone	• CVA • Hyperprolactinemia (max.)
Olanzapine	CVA
Atypical antipsychotic	Weight gain
Ziprasidone	• Weight neutral • Cardiotoxicity

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Drug	Effect
Aripiprazole	• Weight neutral • ↓ Sedation (Least)
Amisulpride	• ↓ Sedation • Hyperprolactinemia
Clozapine	• ↓ EPS (Least) • ↓ Hyperprolactinemia
Paliperidone	Hyperprolactinemia (max.)
3 rd generation antipsychotics	↓ Hyperprolactinemia

Dopamine Pathways :

	Side effect 
mesolimbic	Positive symptoms (Hyperfunctioning of dopamine)
mesocortical	Negative symptoms (Hypofunctioning of dopamine)
Nigrostriatal	EPS
Tuberoinfundibular	Hyperprolactinemia

Extrapyramidal Symptoms (EPS)

00:42:42

Acute EPS :

	Clinical features	Treatment
Akathisia	• Inner restlessness • Lower limb movements • ↑ Suicide risk	β -blockers (m/c : Propranolol)
Acute dystonia	Contraction of group of muscles : • Oculogyric crisis • Tongue protrusion • Trismus • Torticollis & retrocollis • Laryngeal-pharyngeal dystonia • Blepharospasm • Glossopharyngeal dystonia • Opisthotonus	• Anticholinergics • Antihistaminergics

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Chronic EPS :

	Features	Treatment
Chronic akathisia	Akathisia like features for > 6 months	• ↓ Antipsychotic dose • Switch to clozapine (Lesser EPS) • Vesicular monoamine transporter-2 ⊖ (VMAT-2 ⊖) : - Tetrabenazine - Deutetrabenazine - valbenazine (FDA approved)
Tardive dyskinesia 	• Involuntary movement : - Hand wringing - Eye movements • Rabbit syndrome : Perioral tremors	
Tardive dystonia	Contraction of group of muscles	

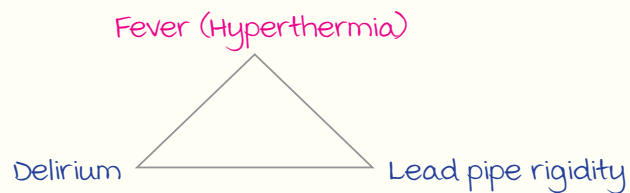
Lethal EPS :

Acute laryngeal dystonia :

New onset breathlessness, stridor.

Neuroleptic malignant syndrome :

- Clinical features :



- Investigations :
 - ↑ Total WBC.
 - ↑ **CPK** (Characteristic).
 - ↑ myoglobinuria.
- Treatment : **BADE**.
 - **Bromocriptine**.
 - **Dantrolene** : DOC.
 - **Amantadine**.
 - **ECT**.

MOOD RELATED DISORDERS

----- Active space -----

mood : Emotion over a period of time.

Affect : Current state of emotion (Cross section).

Depression

00:00:56

Diagnostic Criteria for Major Depressive Episode :

	Core features 🧠	Others
Symptoms	mnemonic : EMI • ↓ Energy • ↓ Mood • ↓ Interest	mnemonic : DIGESCAPS • Depressed affect • Interest ↓ (Anhedonia) • Guilt (Pathological) • Energy ↓ • Sleep issues (Terminal insomnia) • Concentration issues • Appetite changes • Psychomotor changes • Suicidal behavior
Criteria for diagnosis	2 out of 3 symptoms persistent for ≥ 2 weeks	5 out of 9 symptoms persistent for ≥ 2 weeks

Atypical Depression 🧠 :

Features :

- Hypersomnia.
- Hyperphagia.
- Obesity.
- **Lead**en paralysis.
- Rejection sensitivity.
- Preserved mood reactivity.

Rx : MAO inhibitors > SSRI (**TCA : Ineffective**).

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Causes of Depression :

Chemicals	Cognitions	Conflicts
• ↓ 5-HT (Serotonin) • ↓ Norepinephrine • ↓ Dopamine • ↓ BDNF (Brain derived neurotrophic factor) • ↓ 5-HIAA (5-hydroxy indole acetic acid) : metabolite of serotonin	• Worthlessness • Helplessness • Hopelessness Cognitive errors studied by Aaron Beck	• Grief • Anger turned inwards • Ambivalence • Introjection

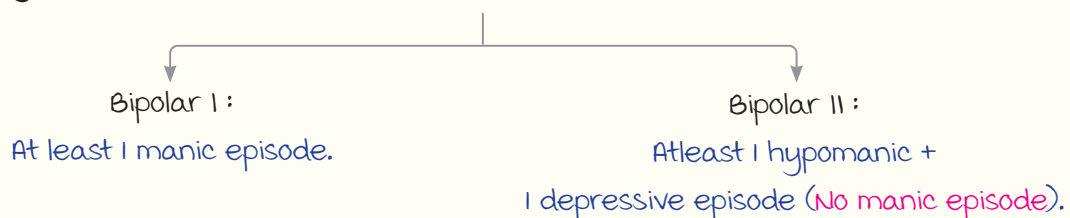
Bipolar Disorder

00:09:20

Features :

- mood/affective disorder.
- Chronic & episodic.
- Characterized by :
 - mania
 - Hypomania
 - Depression

} mixed episode (+)/(-).

Types :Diagnosis of manic Episode :

mnemonic : DIGFAST.

- Distractibility.
- Impulsivity.
- Grandiosity.
- Flight of ideas.
- Activity levels ↑.
- Sleep issues.
- Talkativeness.

Criteria for diagnosis :

	Euphoric/elated	Dysphoric/irritable
No. of symptoms	3 out of 7	4 out of 7
Duration of symptoms	≥ 7 days	

Hypomania vs. Mania :

----- Active space -----

	Hypomania	mania
Duration for diagnosis	≥ 4 days	≥ 7 days
Severity of symptoms	mild	moderate to severe
Psychotic symptoms	$\ominus$	$\oplus/\ominus$
Inpatient care	Not required	might be required
Thought stream disorder	Proximity (Ordered flight of ideas)	Flight of ideas
Seen in	Bipolar I & II	Bipolar I only
Functioning ability	N/ $\uparrow$	Prominent dysfunction $\oplus$

Rapid Cycling Disorder 🧠:Bipolar disorder with ≥ 4 mood episodes/year.

Risk factors :

- Female gender.
- Hypothyroidism.
- Substance use disorder.
- Antidepressants without cover of mood stabilizers.
- Bipolar type II.
- Cyclothymia.
- Hyperthymic personality.

DOC 🧠 : Divalproate.

Persistent Mood Disorders 🧠

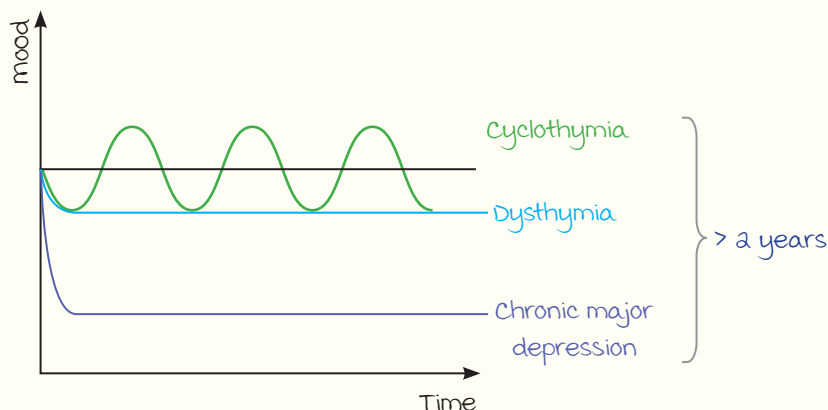
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mood disorders persistent & continuous for > 2 years.

Dysthymia : Chronic, low grade depression.

Chronic major depression : Persistent, severe depressive symptoms.

Cyclothymia : Chronic mood swings b/w mild depression & hypomania.



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Postpartum Mood Disorders

00:22:28

Postpartum period : Following delivery

- upto 4 weeks (DSM).
- upto 6 weeks (ICD).

	Postpartum blues/ baby blues	Postpartum psychosis	Postpartum depression
Onset	2 - 3 days after delivery	Within 1 - 2 weeks (m/c 3 - 7 days)	2 - 6 weeks (upto 1 year)
Duration	< 2 weeks	Variable	> 2 weeks
Symptoms	• Tearfulness • mood swings • Anxiety	• Delusions • Hallucinations • Agitation	• Low mood • Anhedonia • Guilt • Fatigue
Severity	mild	Severe : Psychiatric emergency	moderate to severe
Risk of harm	None	High risk to self & infant	may include suicidal ideation
Rx	Reassurance & support	Hospitalization + antipsychotics ± antidepressants ± mood stabilizers	• Brexanolone • Zuranolone } Neurosteroids
Prevalence	70 - 80% of new mothers	~ 0.1 - 0.2%	~ 10 - 15%

Note :

SSRIs in pregnancy :

- Safe in pregnancy & lactation period : **Sertraline**.
- Avoided in pregnancy :
 - **Fluoxetine** : Long $T_{1/2}$ → No withdrawal symptoms.
 - **Paroxetine** : Short $T_{1/2}$ Abrupt stoppage → withdrawal symptoms.

Suicidal Behaviour

00:27:15

Number of people attempting suicide is 10 to 25 times the rate of suicide.

Key mental Health Conditions a/w Suicidal Behaviour :

Condition	Risk of suicidal behaviour
mood disorders	15%
Schizophrenia	10 - 12%
Substance use disorder	10%
Anorexia nervosa	
Borderline personality disorder	

- Presence of mental illness
 - untreated mental illness
 - Post discharge
- } → ↑ Risk of suicide.

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Risk Factors for Suicide :

- Previous suicide attempt :
Strongest predictor.
- mental illness.
- Hopelessness.
- Access to lethal means.
- unemployment.
- Social isolation.
- male gender (m : F = 2 : 1).

Interventions for mx :

Severe suicidal ideas :

- ECT.
- Ketamine

Rx to ↓ suicidal behaviour 🧠 :

- Lithium.
- Clozapine.

Parasuicide 🧠 :

- AKA DSH (Deliberate self-harm)/NSSI (Non-suicidal self injurious behaviour).
- Self-harm behaviour with no intention to die.

Paradoxical Suicide 🧠 :

↑ Suicidal behaviour during early phase of Rx with antidepressants.

High risk group : Adolescents & young adults.

Cause :

Antidepressants → Change in energy, agitation & ideas → ↑ Suicidal behaviour.

mx :

Continue antidepressants :

- monitor.
- ↓ Dosage/change medication.

Electroconvulsive Therapy (ECT) :

- One of the safest Rx in psychiatry.
- modified ECTs mandatory.

Anaesthetic agent of choice 🧠 : methohexital .

Relative contraindication : ↑ ICT.

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mode :

Bifrontotemporal ECT (m/c) → Electrodes placed 2 - 4 cm above the midpoint b/w tragus & outer canthus of eye.

U/L ECT : Administered on non-dominant side.

Indications :

- Severe mood disorders.
 - Catatonia.
 - Neuroleptic malignant syndrome.
 - Suicidal patients.
 - Resistant seizures
 - Resistant Parkinson's disease
- } Less commonly used.

Treatment Resistant Depression

00:35:52

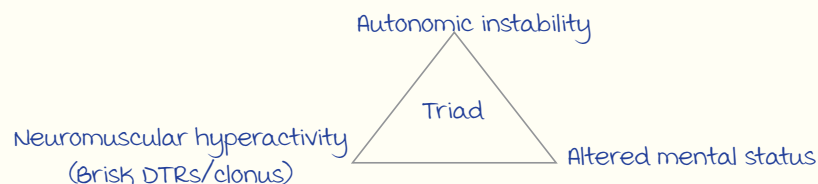
Prevalence : 10 - 20%.

Augmentation strategies :

- Esketamine.
- Adding low dose of :
 - Antipsychotics.
 - Lithium.
 - Thyroxine.
 - Folic acid.
- Somatic Rx :
 - ECTs.
 - rTMS (Repetitive trans magnetic stimulation).
 - VNS (Vagus nerve stimulation).
 - Light therapy : Rx of choice for seasonal affective disorder.
- Combining antidepressants (Risk of serotonin syndrome).

Serotonin Syndrome :

Symptoms :

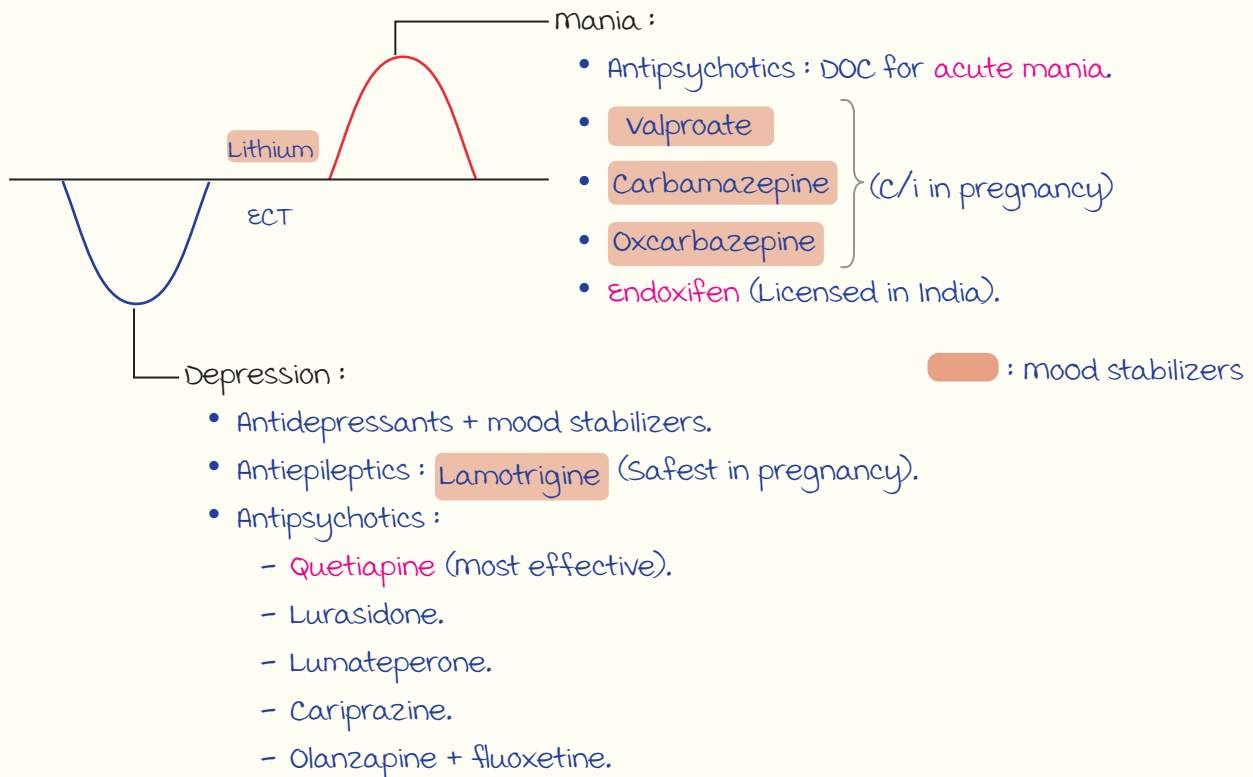


Rx 🧠 : Cyproheptadine.

Pharmacological Management of Bipolar Disorder

00:38:22

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Lithium Therapy

00:41:13

Side Effects :

Long term effects :

- Diabetes insipidus nephropathy.
- Hypothyroidism.

Tremors :

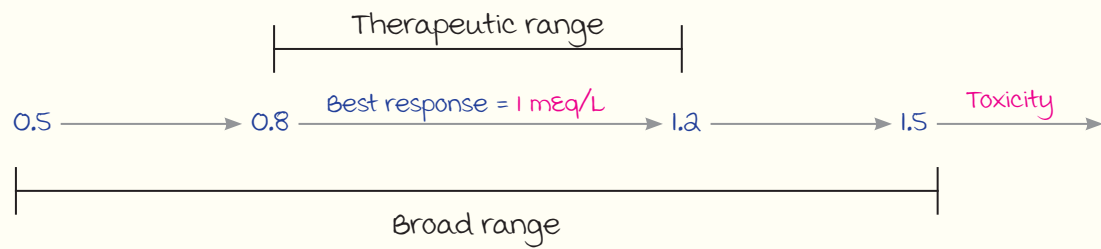
- Benign fine tremor :
Rx by β blocker (Propranolol)/anticholinergics (Trihexyphenidyl).
- Coarse/malignant tremors : Sign of lithium toxicity.

Lithium Toxicity :

Li level $> 1.5 \text{ mEq/L}$:

- myoclonic jerks.
- Coarse tremors.
- Nystagmus.
- Ataxia.
- Seizures.
- Incontinence.

----- Active space -----

Lithium Levels (mEq/L) :Steady state of Li  :

- Attained in 5 days ($T_{1/2}$: 24 hours).
- Blood sample collected after 10 - 12 hours fasting following last dose.

ANXIETY, STRESS AND OTHER DISORDERS

----- Active space -----

Anxiety Disorders

00:00:34

Abnormal anxiety :

- Excessive.
- Persistent.
- Irrational.
- Interferes with functioning.

Generalized Anxiety Disorder (GAD) :

- m/c anxiety disorder.
- Excessive, uncontrollable worry about multiple aspects.

c/f :

- Free floating anxiety.
 - Irritability.
 - motor tension/Restlessness.
 - Sleep disturbances.
 - Inability to relax.
- } ≥ 6 months.

Rx  :

- SSRIs/SNRIs (m/c).
- Benzodiazepines : Cautious use (D/t abuse potential).
- Relaxation techniques :
 - Jacobson progressive muscle relaxation (JPMR) technique.
- Buspirone :
 - Non benzodiazepine anxiolytic.
 - SHT_{1a} partial agonist.
 - No sedation/abuse potential.
 - Safe for long term use.
 - Longer duration for onset of action.

Panic Attack vs. Panic Disorders

00:03:58

Panic attack	Panic disorder
• Sudden, severe anxiety episodes • Lasts 5 - 15 mins • Autonomic arousal : - Sweating - Dryness of mouth - Tremors of hand - Feeling of impending doom	• Recurrent de novo panic attacks • Anticipatory anxiety • Agoraphobia $\oplus/\ominus$

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Panic attack triggers 🧠:

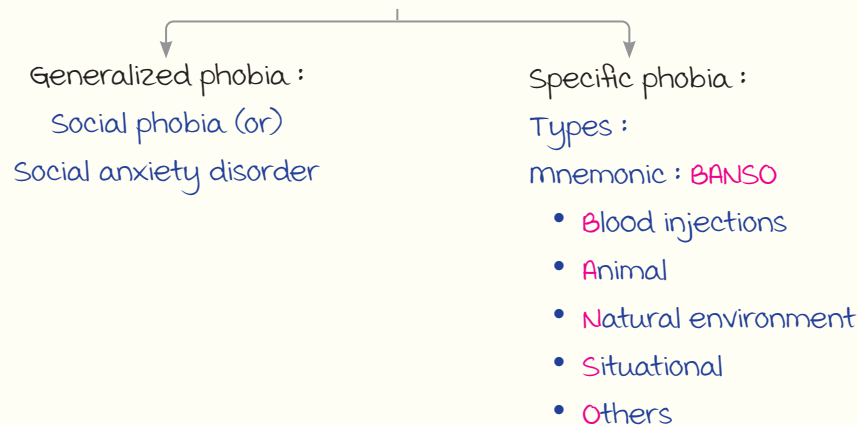
- Caffeine.
- Cannabis.
- Cocaine.
- Cholecystokinin.
- CO₂.
- Nicotine.
- Lactate.
- Theophylline.
- Flumazenil.
- Yohimbine.

Phobia

00:07:03

Excessive & irrational fear.

Types :



Behavioural interventions 🧠:

Relaxation techniques : JPMR, deep breathing.

Cognitive behavioural therapy (CBT) : Restructuring anxious thoughts.

mindfulness-based stress reduction technique (MBSRT).

Exposure therapy : Gradual confrontation of fears.

Flooding : Immediate & full exposure to the most feared stimuli.

Systematic desensitization : Relaxation + gradual & graded exposure to feared stimuli.

Interoceptive exposure : Deliberate induction of physical symptoms to habituate.

Grief

00:10:36

mental illnesses requiring a causative factor :

- Grief.
- Acute stress reaction.
- Acute stress disorder.
- Adjustment disorder.
- Post traumatic stress disorder (PTSD).

Stages of Grief :

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- Described by **Kubler Ross**.
 - mnemonic : **DABDA**.
 - **Denial/Ostrich** phenomena.
 - **Anger** (On self/others/dead).
 - **Bargaining**.
 - **Depression**.
 - **Acceptance**.
- } All stages need not be present.

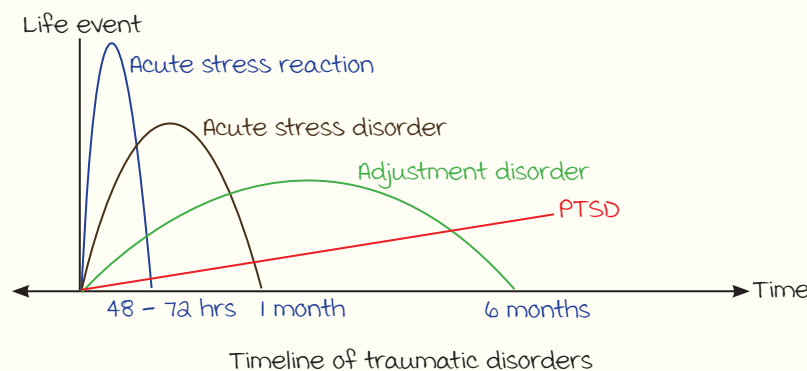
Types of Pathological Grief :

	Description
Chronic/ prolonged	• ≥ 6 months (ICD) • ≥ 12 months (DSM)
Absent	No reaction
Delayed	Reaction delayed by ≥ 2 weeks
Hypertrophic	Intensified emotional response
Anniversary	Reaction around anniversary of the loss
Traumatic	Prolonged + hypertrophic grief

Traumatic Disorders

00:14:15

Types :



	Acute stress reaction	Acute stress disorder	Adjustment disorder	PTSD
Resolution	48 - 72 hrs	≤ 1 month	≤ 6 months	-
Symptoms	Shock/ dream-like/ dazed state	• Autonomic arousal • Startle response • Preoccupied about stress	• AKA situational depression • mild anxiety & depression symptoms • Conduct/behavioural problems	• minimum ≥ 4 wks • Avoidance • Hyperarousal • Nightmares • Flashbacks

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Post Traumatic Stress Disorder (PTSD) :Causative factor : **Life threatening event.**

Risk factors :

Pre trauma factors	Trauma related factors	Post trauma factors
- H/o mental illness Adverse childhood experience - Female gender - Low social support/education - Family h/o psychiatric illness	External locus of control man made disasters > Natural disasters	- Lack of social support - Ongoing stress - maladaptive behaviour - maladaptive appraisals

Delayed onset PTSD : Onset of symptoms > **6 months** after event.

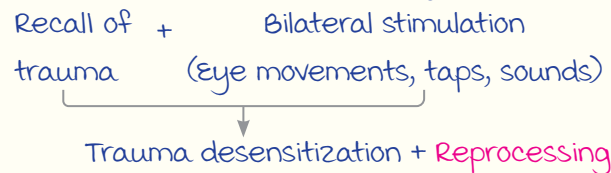
Complex PTSD :

3 key features :

- Affective dysregulation.
- Negative self concept.
- Interpersonal relationship problems.

Rx :

- CBT : Best Rx.
- Eye movement desensitization & reprocessing (**EMDR**) : most specific.

**Obsessive, Compulsive & Related Disorders**

00:24:18

O^a B^a H^a :

- O**bsessive compulsive disorder (OCD)
- O**lfactory reference syndrome
- B**odily focused repetitive behaviour disorder
- B**ody dysmorphic disorder
- H**oarding disorder
- H**ypochondriasis

Hypochondriasis :

- AKA Illness anxiety disorder (DSM-5).
- Illness absent/mild illness.
- Preoccupation about having/developing illness : > **6 months**.

Hoarding disorder :

- Independent illness (Earlier : Type of OCD).
- Store items of insignificant value.

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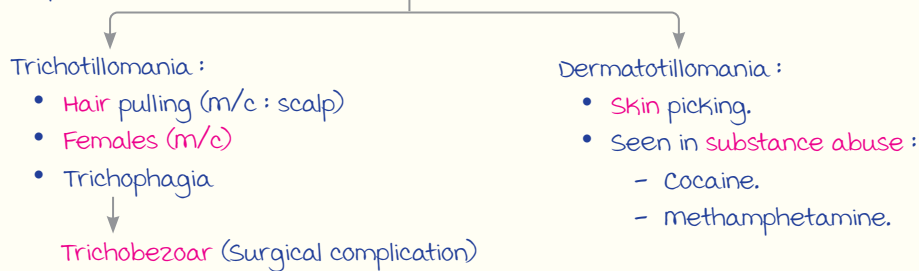
Body dysmorphic disorder :

Preoccupation with body shape/colour/size : > 6 months.

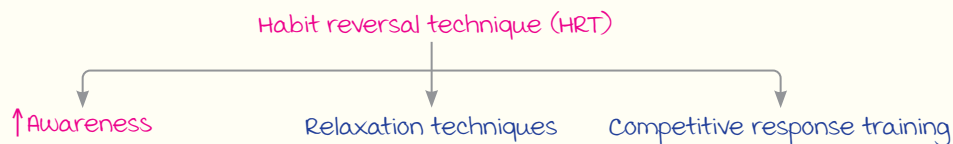
Olfactory reference syndrome : Belief that bad smell emanates from their body.

Bodily focused repetitive behaviour disorder :

- Compulsive behaviour :



- Rx :



Obsession vs. Compulsions :

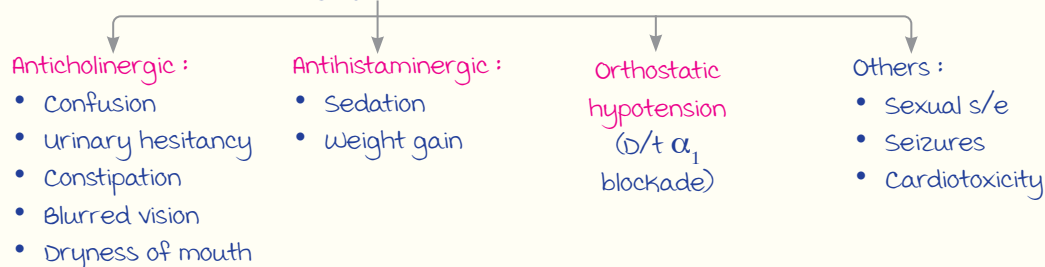
Obsessions	Compulsions
• Repetitive • Intrusive • Distressing • Ego dystonic	• Repetitive • Disabling • Distressing • Ego-dystonic
Thoughts, images, urges ↓ ↑ Anxiety	Physical/mental acts ↓ ↓ Anxiety

management of OCD :

Pharmacological Rx 🧠 :

SSRI₁ → SSRI₂ → Clomipramine (TCA) :

- Anti-obsessive property.
- TCA toxicity : Sodium bicarbonate Rx.
- S/e :

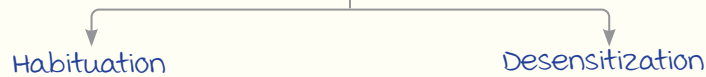


----- Active space -----

Non-pharmacological Rx :

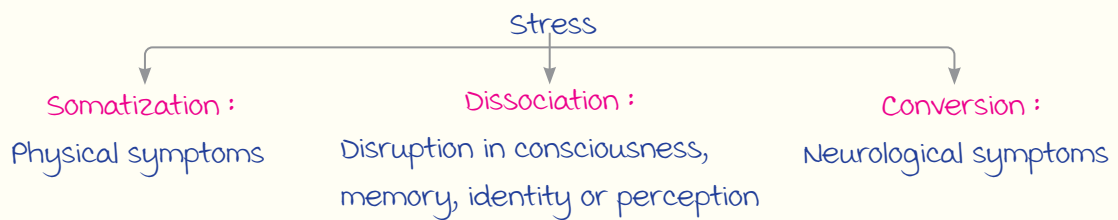
- Repetitive transcranial magnetic stimulation (rTMS) : used in resistant OCD.
- Exposure response prevention (ERP) :

2 principles :



Somatization, Dissociation and Conversion Disorders

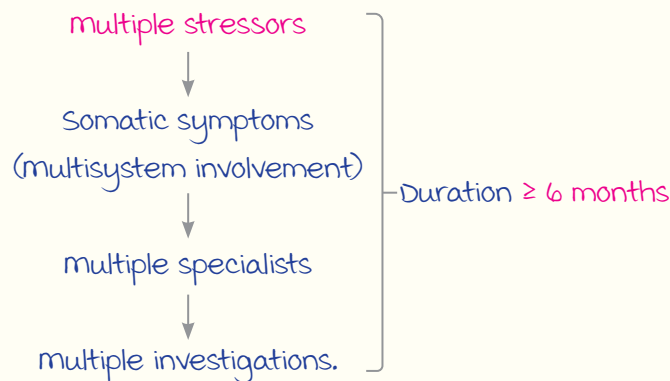
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Somatic Symptom Disorder :

- AKA Bodily distress disorder (ICD).
- Earlier name : Somatization/Somatoform disorder/Briquet Syndrome.

Features :



Rx : Re-attribution therapy (↓ Stress → ↓ Symptoms).

Dissociative Disorders :

Dissociative amnesia : Circumscribed loss of traumatic memory.

Dissociative fugue :

- Loss of primary identity + purposeful wandering.
- Each identity retained for weeks/months.

Dissociative identity disorder/multiple personality disorder :

≥ 2 identities/alters; Frequent shifts between them.

Depersonalization : Detachment from self.

Derealization : Detachment from surroundings.

Conversion Disorders

----- Active space -----

- Old term : Hysteria
- New terms :

	AKA	Group
DSM-5	Functional neurological symptom disorder	Somatic symptom disorders
ICD-II	Dissociative neurological symptom disorder	Dissociative disorders

Pathophysiology :



La Belle indifference : Lack of concern about serious symptoms.

Ganser syndrome :

- Prison/forensic settings (Prison psychosis).
- 40 - 50% patients : underlying organic conditions.
- c/f :
 - Vorbeireden : Approximate answers.
 - Hallucinations.
 - Confusion.
 - Conversion symptoms.

Factitious Disorders

00:46:04

- AKA Munchausen syndrome/Doctor shoppers.
- Deliberate creation of symptoms for medical attention.
- Not conscious secondary gain.

c/f :

Proxy : Brings dependent with concerns/complaints.

Peregrination : Travel long distances for consultation.

Pseudologia fantastica : Excessive pathological lying.

Grid abdomen : Multiple scarred anterior abdominal wall.

Note :

malingering : Create symptoms for conscious secondary gain.

EATING, SLEEP AND OTHER DISORDERS

Eating Disorders

00:00:32

SCOFF Questionnaire :

Rapid screening tool for eating disorders.

- **S** : Do you make yourself **s**ick because you feel uncomfortably full?
- **C** : Do you worry you have lost **c**ontrol over how much you eat?
- **O** : Have you recently lost over **o**ne stone (6.35 Kgs) in a 3 month period?
- **F** : Do you believe yourself to be **f**at when others say you are too thin?
- **F** : Would you say that **f**ood dominates your life?

≥ 2 Yes → Further screening.

Types of Eating Disorders :

	Binge eating disorder (m/c)	Anorexia nervosa	Bulimia nervosa
Onset	Young adults	-	-
BMI	Overweight/obese	< 18.5	Normal
c/f	3/5 ⊕ : • B : Eating beyond fullness • I : Eating in isolation • N : Eating when not hungry • G : Guilty after eating • E : Eating rapidly	• Types : - Restrictive - Binge purge • Fear of weight gain • Distorted body image • 10% suicide risk	• Binge eating • Purging • Compensatory mechanisms ⊕
Complications	-	• Amenorrhea • Refeeding syndrome : ↑ Surge in calories ↓ - Compensatory hyper-insulinemia ↓ - Electrolyte imbalance : ↓ PO_4^{3-}, ↓ Mg^{2+}, ↓ K^+ ↓ Cardiac arrhythmias & death	• Retching : - Loud sounds while vomiting - Feature of psychological vomiting • Russell sign : ulceration & pigmentation on the dorsum of the hand • mallory weiss syndrome : Lower esophageal tear
Rx	L isdexamfetamine	• Olanzapine • Fluoxetine • mood stabilizers	Fluoxetine

EEG Waves

00:09:53

----- Active space -----

Stage		Waves	Frequency
Awake	Open eye, active & thinking	β waves (Frontal & central areas)	≥ 14 Hz
	Closed eye & relaxed, meditation	α waves (Occipital areas)	7 - 14 Hz
N_1 : • Light sleep, microsleep • Hypnic jerks • Slow rolling eye movements		θ waves	4 - 7 Hz
N_2 : • Sleep spindles : - Burst of α waves - $\uparrow$ Frequency - $\downarrow$ voltage  N_3 : • Bruxism 		• Sleep spindles : - Burst of α waves - $\uparrow$ Frequency - $\downarrow$ voltage 	12 - 14 Hz
		• K-complex : - Biphasic waves - $\downarrow$ Frequency - $\uparrow$ voltage 	-
N_3 : • Restorative/delta sleep/ slow wave sleep • most difficult to awaken		Delta waves	< 4 Hz
REM : Paradoxical sleep		• β waves • Saw tooth appearance of waves 	≥ 14 Hz

Sleep Disorders

00:15:53

Types :

	Parasomnia	Dyssomnia
Definition	Abnormal behaviours during sleep	Sleep quality/quantity issues
Effect on sleep	Sleep is normal, disrupted by events (walking, talking)	Difficulty falling/staying asleep, excessive sleepiness
Examples	- Sleepwalking - Night terrors - Nightmare disorder - REM behaviour disorder	- Narcolepsy - Insomnia - Hypersomnia - Jet lag syndrome

----- Active space -----

Night Terrors vs. Nightmares :

	Night terrors (Pavor Nocturnus)	Nightmares
Stage of sleep	N ₃	REM
Amnesia	⊕	⊖
Autonomic arousal	⊕	⊖
Post arousal confusion	⊕	⊖
Age group	Adolescents & younger children	Any age

Narcolepsy :

Deficiency of hypocretin/orexin.

Types :

- Type 1 (Classical) : Cataplexy + ↓ hypocretin.
- A/w HLA DQB 06-02.
- Type 2.

C/F :

- Sleep attacks (REM).
- Sleep paralysis.
- Cataplexy : Extreme emotional stimuli
↓
Hypotonia.
- Hallucinations : Hypnagogic/hypnopompic.
- ↓ Sleep latency.
- SOREM : Sleep onset REM in routine EEG.

Rx :

- modafinil.
- Armodafinil.
- Solriamfetol.
- Pitolisant.
- Lis-dexamfetamine.
- methylphenidate.
- Amphetamine salts.

Cataplexy Rx :

- Sodium oxybate.
- Pitolisant.

Personality Disorders

00:23:24

Big 5 personality traits (Big 5 model/Five factor model) 🧠 :

mnemonic : OCEAN.

- Openness.
- Conscientiousness.
- Extraversion.
- Agreeableness.
- Neuroticism.

Clusters :

----- Active space -----

Cluster	Personality disorder	Features
A (Odd/eccentric)	Paranoid	Distrust & suspicion of others
	Schizoid	Detached & emotionally cold
	Schizotypal	- Eccentric, odd & aloof - Prone to develop schizophrenia
B (Dramatic, emotional, erratic)	Borderline 	- Emotional instability - Deliberate self harm (DSH) - Prone to develop bipolar disorder
	Histrionic	Attention seeking
	Narcissistic	- Self admiration - Grandiose - Abusive & manipulative
	Antisocial 	- Lack empathy - Cannot follow law & regulations - Prone to addiction disorders
C (Anxious & fearful)	Anankastic/ obsessive-compulsive personality (OCPD)	- Orderliness & perfection - Ego-syntonic (OCD : Ego-dystonic) - Prone to depression
	Dependent	Submissive & clinging
	Avoidant	Avoidance

Emotionally unstable personality disorder (EUPD) :

- AKA **Borderline** personality disorder.
- C/f :
 - mood fluctuations.
 - **multiple relationships.**
 - **DSH.**
 - Impulsivity.
- Rx : Dialectical behavioural therapy (**DBT**).

ICD-II updates :Five domains : **ND³A**

- **Negative affectivity.**
- **Detachment.**
- **Disinhibition.**
- **Dissocial.**
- **Anankastia.**

Types :

- mild PD.
- moderate PD.
- Severe PD.

Personality difficulty :

- Individuals with problematic personality traits.
- **Do not meet criteria** for personality disorders.

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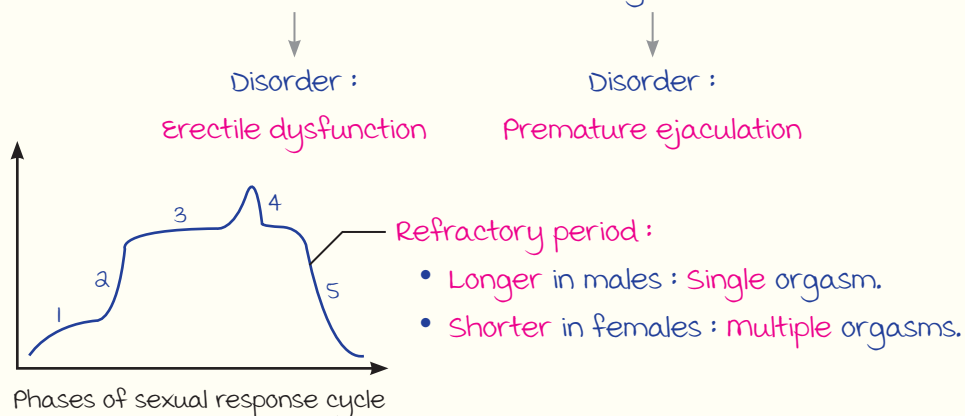
Sexual Disorders

00:30:14

Phases of Sexual Response Cycle 🧠:

mnemonic : **DEPOR**

Desire → Excitement → Plateau → Orgasm → Resolution.

**Rx :**

Disorder 🧠	Treatment
Erectile dysfunction	- Phosphodiesterase $\ominus$: - MOA : $\uparrow$ Nitric oxide & cGMP levels - E.g. : - Tadalafil - Sildenafil - Vardenafil - Avanafil - PGE analogue : Alprostadil - Short acting - muse : - medicated urethral system for erection - used for 1x & Rx.
Premature ejaculation	- Squeeze technique (most effective) - Start stop technique - Sensate focusing (Emphasize on non consummation aspects)

Sexual Paraphilias :

Deviation from normal consensual adult behaviour.

	Cause for arousal
Voyeurism	- Watching unsuspecting individuals undress/have sex - Diagnosed in > 18 yrs only
Exhibitionism	Exposing genitals to unsuspecting individual
Frotteurism	Touching non-consenting individuals
Pedophilia	Sexual interest in pre-pubescent children (< 18 yrs)
Sexual sadism	Arousal from non-consensual suffering
Sexual masochism	Arousal from suffering, humiliation or pain
Fetishism	Arousal from inanimate objects/non-genital body parts
Transvestic fetishism	Arousal from cross dressing

Note :

Cross dressing :

- Seen in gender dysphoria/ incongruence.
- No sexual arousal.

Substance Use Disorders

00:39:40

----- Active space -----

Spectrum of Substance Use :

	Description
Teetotaler	Do not use alcohol
Recreational/social	use for positive effects
Abuse	use to feel normal
Hazardous use	Potential for damage
Harmful use	usage despite actual harm
Dependence/addiction	• Craving • Loss of control • Compulsive use • Central preoccupation (Salience) • Continued use despite consequences • Tolerance • Withdrawal symptoms

} ≥ 3 ⊕

Stages of motivation/Change :

Given by Prochaska & DeClemente.

Precontemplation : Denial, non-acceptance.

Contemplation : Acceptance.

Decision/planning : Preparedness.

Action : Overt change.

Maintenance : Persistence > 6 months.

Recovery/relapse : 100% self efficacy, 0% temptation.

medical use of Cannabis :

Components :

- Δ-9 tetrahydrocannabinol : Psychoactive (Causes dependence).
- Cannabidiol : medical use.

Synthetic cannabis-like drugs 🧠 :

- Dronabinol.
- Nabilone.

Indications :

- Epilepsy (Dravet and Lennox-Gastaut) : DCGI approved.
- Chronic pain syndromes.
- Multiple sclerosis.
- Movement disorders.
- Cachexia (D/t cancer/HIV/chemotherapy).
- Glaucoma.
- ADHD.
- Anxiety disorders.

----- Active space -----

Alcohol Withdrawal :

Timeline & symptoms :

Time since last drink	Symptoms 
6 - 12 hrs	Simple withdrawal : • ↓ Sleep/insomnia • Tremors • Tachycardia • ↑ Sweating • Rebound REM phenomena
12 - 24 hrs	Perceptual hallucinations : • Auditory (m/c) • Visual
24 - 48 hrs	GTCS
48 - 72 hrs	• Delirium tremens • Lilliputian hallucinations

Delirium tremens vs. Perceptual hallucinations :

	Delirium tremens	Perceptual hallucinations
Time of onset	48 - 72 hrs	12 - 24 hrs
Sensorium	Altered	Clear
m/c hallucination	Visual	Auditory

Wernicke's encephalopathy vs. Korsakoff psychosis :

	Wernicke's encephalopathy	Korsakoff psychosis
Cause	Thiamine deficiency	
Onset	Acute/emergency	Chronic
c/f	Mnemonic : G OA • Global confusion • Ophthalmoplegia • Ataxia	• Confabulation • Anterograde > Retrograde amnesia
Lesion	Petechial hemorrhages in : • mammillary bodies • Periaqueductal grey matter	• Anterior thalamic nuclei • medial mammillary nuclei

Rx of Alcohol Use Disorder :

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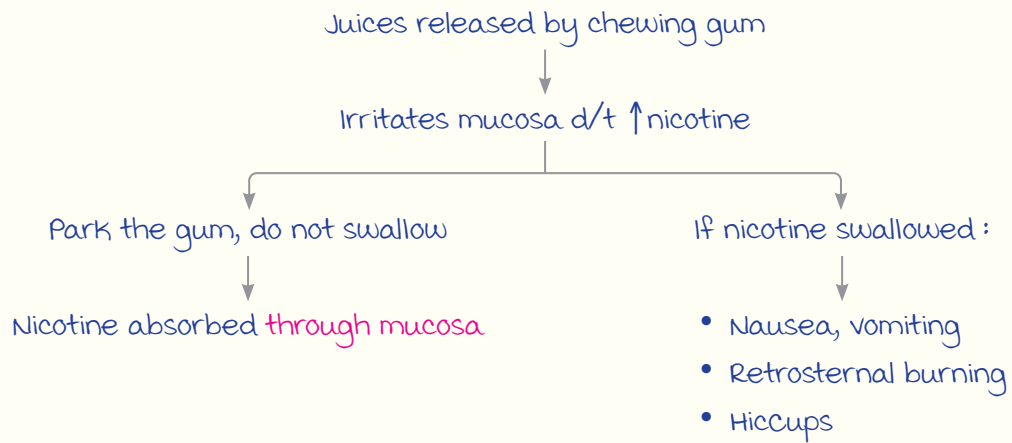
	Drugs
Withdrawal Rx	Benzodiazepines : • Lorazepam • Oxazepam • Temazepam (most important) } • mnemonic : LOT • Elderly, liver dysfunction
Anti-craving Rx	mnemonic : ANTSOB • Acamprosate • Naltrexone • Topiramate • SSRIs • Ondansetron • Baclofen
Deterring/ aversive drug	Disulfiram : • Irreversible aldehyde dehydrogenase ⊖ • ↑ Aldehyde, histamine, serotonin, dopamine • used in motivated patients • Avoided in : - Psychosis - Peripheral neuropathy • Started 12 - 24 hrs after last drink • Avoid alcohol for 10 - 14 days after stopping

management of Nicotine Use Disorders :

	Drugs
Anti-craving Rx 	• Bupropion : ↑ Nor-epinephrine & dopamine • Varenicline : - $\alpha_4\beta_a$ partial agonist - most effective anticraving drug
Nicotine replacement therapy	• Forms : - Gums } 2 mg, 4 mg - Lozenges } - Patch : 7 mg, 14 mg, 21 mg • Advantage : ↓ Exposure to other harmful organic compounds

----- Active space ----- Precautions for using gums 🧠 :

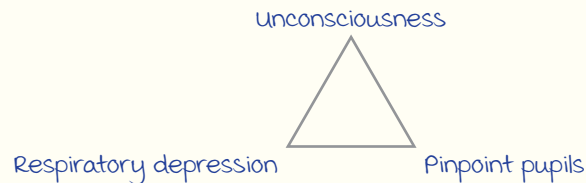
- Chew & park method :



- Avoid hot food & beverages for 15 - 20 mins (↓ Absorption).

Opioid Poisoning :

Triad :



mx :

	Drugs
Overdose	• IV Naloxone • Naltrexone : - Long acting antagonist - Oral drug (Not used in emergencies)
Withdrawal Rx	• Clonidine : Controls autonomic symptoms • Buprenorphine : - Partial agonist • methadone : - Full agonist - Dependence & abuse potential
Maintenance Rx	• methadone • Buprenorphine • Buprenorphine + Naloxone : Preferred - Naloxone : μ receptor antagonist - ↓ Abuse potential

NEUROCOGNITIVE, NEURODEVELOPMENTAL AND OTHER DISORDERS

----- Active space -----

Delirium

00:00:39

- m/c organic/neurocognitive disorder.
- AKA ICU psychosis/acute organic brain syndrome.

c/f:

Mnemonic A³S³:

- Acute onset.
- Altered sensorium.
- Autonomic dysfunction.
- Sleep reversal.
- Sundowning.
- Slowing of EEG waves (Generalized).

Bedside Assessment of Delirium :

Confusion assessment method (CAM).

Mini mental status examination (MMSE) :

- Total points = 30 (≤ 24 : Significant).
- 5 domains evaluated :
 - Orientation.
 - Registration.
 - Attention → Serial subtraction test (100 - 7).
→ Spell 'WORLD' in reverse order.
 - Recall.
 - Language.
- Drawback : Biased towards dominant lobe.

Common Altered Mental States :

	Features
Twilight state	• Abrupt discontinuity in consciousness • maybe a/w behavioural & emotional outburst.
Oneiroid state	• Dream-like confused state • A/w hallucinations
Stupor	• AKA akinetic mutism • minimally responsive alert state

Delirium vs. Dementia :

	Delirium	Dementia
Onset	Acute (48 - 72 hrs)	Chronic (> 6 m)
c/f	Altered sensorium	• Clear sensorium • Progressive cognitive decline

----- Active space -----

Dementia

00:09:42

Types :

	Subcortical dementia	Cortical dementia
e.g.	• Parkinson's disease • Supranuclear palsy • Huntington's chorea	• Alzheimer's disease (m/c) • Pick's disease • Creutzfeldt Jakob disease
c/f	• motor problems • movement problems • memory problems	• Aphasia • Apraxia • Agnosia

Reversible causes :

- Vit **B₁₂** deficiency.
- **Hypothyroidism**.
- Early stages of :
 - Neuroinfection.
 - Head injury (**SDH** : Alcoholics).
 - Normal pressure hydrocephalus (**NPH**).

Pseudodementia :

- Seen in **elderly** patients with **severe depression**.
- 'Don't know' answers.
- Rx : Antidepressants.

Alzheimer's Disease :

Characteristic features :

Risk factors  :

- Advancing **age** (Strongest).
- Apo lipoprotein **E-4**.
- **Presenilin 1 & 2** → Familial autosomal dominant conditions.
- Down's syndrome.
- Females.
- Diabetes, hypertension, dyslipidemia.
- Lower education status.
- Head injury.
- minimal physical activity.

Rx :

Drug class	Example	Stage of disease
Acetylcholinesterase inhibitors	Donepezil	All cases
	• Rivastigmine • Galantamine	mild to moderate
NMDA antagonists	memantine	Severe
monoclonal antibodies	• Aducanumab • Lecanemab • Donanemab	Early

Lewy Body Dementia 🧠 :

Intracellular accumulation of α -synuclein.

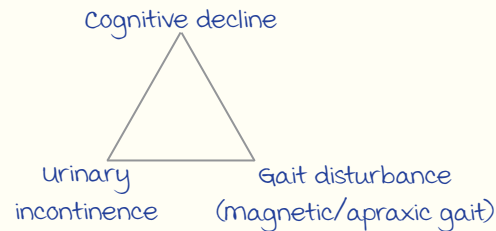
Characteristic features :

- Fluctuating course of illness.
- Prominent visual hallucinations.
- Extrapyraximal symptoms :
 - Even with ↓ antipsychotic dose.
 - Clozapine can be used.
- REM sleep behaviour disorders.
- Capgras syndrome.

Normal Pressure Hydrocephalus :

Reversible in early stages.

Triad : Wet, wacky, wobbly.



----- Active space -----

Neurodevelopmental Disorders

00:19:23

Intellectual Developmental Disorder (DSM-5) :

- AKA Disorder of intellectual development (ICD-11).
- Earlier term : mental retardation.

c/f :

Delay in :

- Intellectual functioning.
- Adaptive behaviour.

Salient aspects :

Severity	IQ	Disability %	maximum functioning age
mild	50 - 69	50%	12 yrs
moderate	35 - 49	75%	9 yrs
Severe	20 - 34	90%	6 yrs
Profound	< 20	100%	3 yrs

Autism Spectrum Disorders :

m/c in males.

c/f :

- | | | |
|--|---|--|
| <ul style="list-style-type: none"> • Stereotypic behaviours :
Restricted, repetitive behaviours. • Socialization deficits : <ul style="list-style-type: none"> - Poor eye to eye contact. - Communication difficulties. | } | <p>Core features.</p> <ul style="list-style-type: none"> • Speech problems. • 50% comorbid with IDD. |
|--|---|--|

----- Active space -----

Rett syndrome :

- m/c in females.
- Autosomal dominant X-linked gene (*MECP-2*).
- c/f :
 - Normal growth for 6 months - 2 yrs → Regression of milestones.
 - midline stereotypic behaviours.
 - microcephaly.
 - Loss of hand abilities.
- Rx : Trofinetide (Synthetic analogue of IGF-1).

Attention Deficit Hyperactivity Disorder :

40 - 50% children with ADHD → Adults with ADHD.

Core symptoms :

- Attention problems.
- Hyperactivity.
- Impulsivity.

Diagnostic criteria :

- Onset : ≤ 12 years of age.
- Duration : ≥ 6 months.
- Present in ≥ 2 settings.
- 6 attention symptoms + 6 hyperactivity/impulsivity symptoms.

Rx :

Stimulants (more effective)	Non-stimulants
• methylphenidate • Dextroamphetamine • Lisdexamfetamine • mixed amphetamine salts	• Atomoxetine • Viloxazine • Clonidine • Guanfacine
} Not available in India	} Norepinephrine modulation } α_a agonists

HEADSS Assessment :

Psychosocial interview tool for adolescents :

- Home environment.
- Education/employment.
- Activities (Hobbies/sports).
- Drugs.
- Sexuality.
- Safety.
- Suicide/self harm.

Oppositional Defiant Disorder (ODD) vs. Conduct Disorder (CD) :

----- Active space -----

	ODD	CD	Antisocial personality disorder
Onset	< 10 yrs	> 10 yrs	≥ 18 yrs
Duration	≥ 6 months		
Defiance, disobedience	⊕		
Deceitful	⊖	⊕	
Destructive	-	To people, property & animals	

Adolescent onset ODD : Symptoms start in adolescence.

Concepts Related to Mind

00:35:38

models of mind :

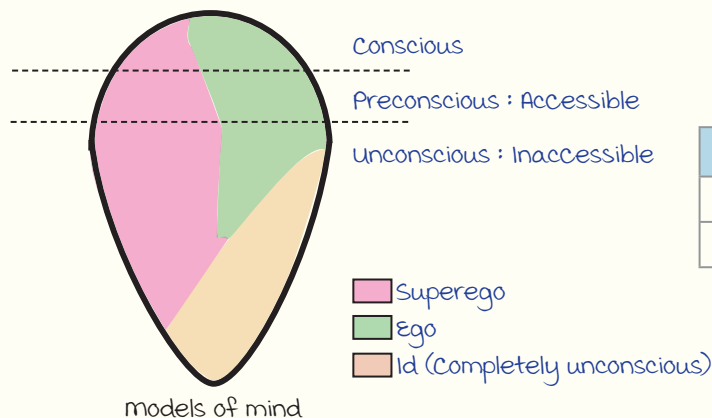
Proposed by Sigmund Freud.

Structural :

- Id.
- Ego.
- Superego.

Topographical :

- Conscious.
- Preconscious.
- Unconscious.



	Id	Ego	Superego
Onset	Birth	4 - 6 months	2 - 3 years
Principle	Pleasure	Reality	morality

Defense mechanisms :

Produced by ego.

mature defense mechanisms : HAAASS.

- Humour.
- Anticipation.
- Altruism.
- Asceticism.
- Sublimation.
- Suppression : Forgive & forget (Conscious defence mechanism).

Repression :

- m/c defense mechanism.
- Forgive but do not forget.

----- Active space -----

Projective Tests 🧠:uses **ambiguous** stimuli.

1. Rorschach's inkblot test :



2. Thematic apperception test :



3. Sentence completion test.

4. Draw a person test.

Doctor Patient Relationship 🧠:Redirection of **+ve** or **-ve** feelings towards a particular person.Catharsis vs. Abreaction :

	Catharsis	Abreaction
meaning	ventilation/sharing/expression	Abnormal reaction
Benefit in therapy	⊕	⊕/⊖

Mental Health Care Act (MHCA) 2017

00:51:55

Prohibited Procedures :

- Seclusion/chaining.
 - Sterilization procedures.
 - **unmodified** ECT.
 - ECT in minors.
 - Psychosurgery/neurosurgical intervention for mental illness.
- } Done only with mental health review board (MHRB) permission.

MHRB :

- Quasi-judicial body.
- Fines psychiatrists for violation of MHCA.

Advanced Directive :

----- Active space -----

Legally valid document registered under MHRB.

3 components :

- How to be treated.
- How to not be treated.
- **Nominated representative** : Adult (≥ 18 yrs) responsible for taking decisions if one loses capacity of thinking.

Admissions :

Types :



Emergency admissions :

- Only at mental health establishments (MHE).
- In a regular facility $\rightarrow$ ≤ 72 hours only.
 $\rightarrow > 72$ hrs $\rightarrow$ Refer to MHE.

Rights of Persons With Disabilities (RPWD) Act :

Included illnesses :

- Intellectual developmental disorder.
- mental illnesses.
- **Autism spectrum disorder.**
- **Specific learning disorder (DSM)/** } New additions.
 Developmental learning disorder (ICD)

Benchmark disability : $\geq 40\%$.

mental Health Gap (mhGAP) :

- WHO programme.
- **Community & algorithm based.**
- Non-specialists $\rightarrow$ Task sharing.

Aim : **Reduce treatment gap** $\approx 80\%$.

Conditions covered :

MNS :

- **mental health illnesses** :
Depression, anxiety.
- **Neurodevelopmental issues** :
Epilepsy, IDD.
- **Substance use disorders** :
Alcohol, nicotine.

